# Archimedean Invitational



# Codebusters Exam

## Division B 2025-26

**1. (250.00 pts)** Solve this Timed Question, remember that every second counts!

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 3 | 1 | 14 | 1 | 0 | 12 | 5 | 0 | 0 | 5 | 0 | 2 | 6 | 4 | 0 | 9 | 0 | 2 | 20 | 3 | 4 | 7 | 5 | 7 | 9 | 4 | 5 |
| Replacement | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

F M R Z T A Z C W R C J U O J R O W G C F F C V R M X Z Z Q . O F O W F C

V R T W R J T U , F C V R M C G C A X V U R , F C V R

L C S Z X W W O C G X F R , F C M X N R O F S X D R W C S R Y O J J R A R G L R

F M X F Q C T M X N R U O N R Y X G Y U O N R Y B R U U .

**3. (273.00 pts)** The following quote needs to be decoded with the Aristocrat Cipher using a K1 alphabet. What does it say?

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 2 | 2 | 3 | 1 | 5 | 0 | 0 | 9 | 1 | 0 | 0 | 5 | 4 | 0 | 0 | 0 | 0 | 3 | 4 | 4 | 4 | 2 | 13 | 4 | 8 | 2 | 2 |
| Replacement | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| Z | H | L | | E | R | M | V | | X | H | | A | V | | U | H | B | L | W | V | C | | R | Y | Y | | X | E | V | | X | T | Q | V | . | | V | M | V | S | | T | S | | X | E | V |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | . | | | | | | | | | | | |

| X | H | L | D | E | | Q | H | Q | V | S | X | W | , | | Z | H | L | | E | R | M | V | | X | H | | A | V | | U | H | B | L | W | V | C | | R | S | C |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | , | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| I | H | W | T | X | T | M | V | . |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | . |

**4. (131.00 pts)** Decrypt the following cipher text that has been encoded using the Atbash Cipher. What does it say?

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| D | S | R | O | V | | G | S | V | | L | G | S | V | I | H | | K | O | Z | B | V | W | | L | M | | G | S | V | | H | F | I | U | Z | X | V | , | | D | V | | D | V | I | V |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | , | | | | | | | | | | |

| U | L | I | X | V | W | | G | L | | S | R | W | V | | F | M | W | V | I | T | I | L | F | M | W | . |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | . |

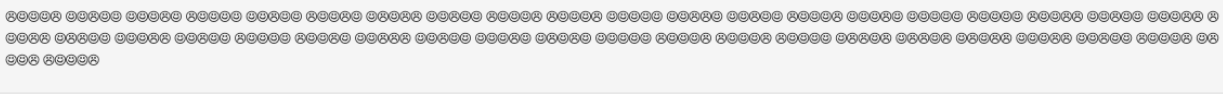
**5. (278.00 pts)** Solve this quote that has been encoded using the Checkerboard Cipher with a polybius key of PLANETARY. What does it say?

SN AV SE SN SU TS SS AV RE SN TE TV AS SN SU RS SV SE SN SU TV RN SN SU SU AS AU AS SE SN TV SS AS SV SV AS TV SS TE

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| A | A | A | A | A | A | A | A | | A | A | A | A | A | A | A | A | | A | A | A | A | | A | A | | A | A | A | A | A | A | A | A | A | | A | A | A | A | A | A | A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

**6. (174.00 pts)**
A quote has been encoded using the Baconian Cipher for you to decode. What does it say?



Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

**7. (153.00 pts)** Solve this quote that has been encoded using the Caesar Cipher. What does it say?

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| V | H | G | L | P | H | Q | W | V | F | R | P | E | L | Q | H | W | R | J | H | W | K | H | U | D | Q | G | F | R | P | S | D | F | W | W | R |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| I | R | U | P | V | H | G | L | P | H | Q | W | D | U | B | U | R | F | N | V | . |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | . |

**8. (294.00 pts)** Decrypt the following cipher text that has been encoded using the Affine Cipher. You are told that W decodes to be U and O decodes to be E. What does it say?

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| P | J | O | N | B | G | R | Q | S | Q | P | Y | P | G | I | P | W | N | Q | H | Q | P | Y | K | Q | L | L | O | D | O | Z | P | W | M | L | L | Y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| Q | Z | V | L | W | O | Z | A | O | Y | G | W | . |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | . |

**9. (274.00 pts)** A phrase has been encoded using the Aristocrat Cipher. What does it say?

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 1 | 0 | 7 | 1 | 6 | 1 | 4 | 2 | 3 | 2 | 8 | 0 | 0 | 2 | 0 | 1 | 2 | 3 | 0 | 0 | 0 | 5 | 5 | 11 | 6 | 6 | 3 |
| Replacement | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| E | N | Q | C | P | P | X | Y | W | E | W | F | K | Y | Q | G | J | C | U | W | X | E | G | J | C | C | W | I | Y | V | V | Z | V | O | I | . |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | . |

| W | X | K | V | C | Q | H | K | U | E | W | H | C | K | G | , | W | X | K | X | Y | U | A | K | U | E | W | E | G | W | C | I | U | K | Y | N |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | , | | | | | | | | | | | | | | | | | | | | |

| W | X | Y | W | Z | D | Z | V | K | . |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | . |

**10. (244.00 pts)** Solve this quote that has been encoded using the Porta Cipher. The deciphered text starts with MOSQUI.. What does it say?

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| N | K | F | D | F | R | E | F | E | S | R | G | E | N | C | N | M | N | B | L | O | C | Q | U | A | K | Q | G | W | W | E | R | E | J | V | A | T | K | J |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| G | S | A | P | R | L | K | D | L | U | R | E |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

**11. (401.00 pts)** A phrase in Spanish has been encoded using the Xenocrypt Cipher. What does it say?

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 6 | 19 | 3 | 10 | 8 | 8 | 2 | 0 | 2 | 10 | 4 | 4 | 5 | 0 | 4 | 0 | 8 | 0 | 1 | 2 | 0 | 2 | 0 | 1 | 5 | 1 | 1 |
| Replacement | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| L | E | B | | C | D | A | Ñ | P | | C | E | B | | U | J | A | D | F | | B | P | | A | Ñ | B | P | J | | P | D | | G | F | Ñ | S | B | F | D | | W | B | I | , |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | , |

| A | E | D | X | K | J | | P | J | M | | Z | B | M | J | M | | C | E | B | F | J | X | | B | P | | S | J | U | J | F | | K | B | | D | F | F | D | X | L | E | B | , |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | , |

| L | E | B | | B | X | A | B | X | K | Ñ | J | | P | D | | P | E | I | | L | E | B | | Y | J | R | | M | B | | K | B | M | D | G | D | F | B | A | B | . |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | . |

**12. (241.00 pts)** [SPECIAL BONUS] Solve this quote that has been encoded using the Cryptarithm Cipher. What do the values 4357912 98312 decode to?

SAND+FADING=SUNSET

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| A | A | A | A | A | A | A | | A | A | A | A | A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | |

**13. (233.00 pts)** Decrypt the following cipher text that has been encoded using the Complete Columnar Cipher. The quote has HISTORI somewhere in it. What does it say?

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| H | N | R | I | H | | T | E | N | U | T | | I | C | E | T | Y | | R | U | R | L | T | | D | E | U | T | E | | A | C | E | W | S | | G | O | F | S | W |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| T | H | N | D | E | | T | O | L | H | A | | H | O | A | I | S | | A | R | E | U | N | | X | R | I | S | H | | S | Y | R | I |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

**14. (262.00 pts)** The following quote needs to be decoded with the Aristocrat Cipher using a K1 alphabet. What does it say?

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 2 | 9 | 1 | 5 | 4 | 0 | 1 | 4 | 0 | 0 | 9 | 5 | 0 | 2 | 0 | 0 | 5 | 0 | 2 | 5 | 3 | 3 | 1 | 8 | 7 | 1 | 9 |
| Replacement | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| P | W | L | | D | B | X | ' | K | | C | Z | | B | R | A | B | S | E | | K | W | | R | B | S | H | . | | S | K | ' | T | | K | U | Z | | W | X | H | P | | N | B | P |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | ' | | | | | | | | | | | | | | | | | | | | | . | | | | ' | | | | | | | | | | | | | | |

| P | W | L | | T | L | D | D | Z | Z | E | | – | P | W | L | ' | A | Z | | X | W | K | | Y | W | X | X | B | | T | L | D | D | Z | Z | E | | B | H | H | | K | U | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | – | | | | ' | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| K | S | V | Z | , | | B | X | E | | S | | G | X | W | N | | K | U | B | K | . |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | , | | | | | | | | | | | | | | | | | . |

**15. (276.00 pts)** Solve this quote that has been encoded using the Affine Cipher with a=15 and b=5. What does it say?

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| E | G | N | P | F | E | V | P | C | F | J | E | V | H | S | H | C | J | O | V | J | Z | V | S | R | E | G | N | P | T | U | D | V | E |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| F | P | P | V | R | S | D | N | S | E | U | T | E | E | H | S | . |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | . |

**16. (266.00 pts)** A phrase has been encoded using the Nihilist Substitution Cipher with a keyword of SLUMBER and polybius key of PEACEFULNESS. What does it say?

76 34 34 63 70 45 57 47 53 53 85 38 35 75 36 43 65 56 67 34 66 66 74 66 74 67 36 56 76 55 63 72 67 35 86 69 46 44 83 69 24

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| A | A | A | A | A | A |
|---|---|---|---|---|---|
| | | | | | |

**17. (284.00 pts)** A phrase has been encoded using the Aristocrat Cipher. What does it say?

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 0 | 5 | 3 | 1 | 4 | 1 | 0 | 0 | 5 | 1 | 4 | 6 | 0 | 3 | 0 | 2 | 4 | 7 | 1 | 0 | 6 | 0 | 1 | 4 | 10 | 12 | 1 |
| Replacement | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| Q | K | X | F | Y | B | K | X | W | X | L | X | Q | V | L | X | B | Q | K | Y | W | O | L | P | Y | T | X | C | B | E | Y | I | W | E | L | P |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| R | X | B | Y | T | B | E | I | I | , | N | Y | C | X | T | Y | I | X | L | P | , | N | Y | T | Z | I | Y | W | N | D | E | Q | . | L | Y | J | Y | T |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | , | | | | | | | | | | | , | | | | | | | | | | | | . | | | | | |

| C | X | T | P | Y | Q | Q | K | O | Q | . |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | . |

**18. (322.00 pts)**
Solve this Fractionated Morse. You are told that the quote has COLLEC in it corresponding to the encoded text Z G W K A R N D G X, meaning A = −••; D = •x−; G = •−•; K = −x•; N = ••x; R = x•−; W = x—; X = x−x; Z = xx−.

B M J O Y J N G U A Z L S R Y G Q V S R V D N N K D J C X H Z M I Z G W K A R N D G X N H V

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

**19. (373.00 pts)** A phrase has been encoded using the Aristocrat Cipher. What does it say?

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 3 | 7 | 3 | 3 | 5 | 0 | 0 | 3 | 1 | 6 | 3 | 15 | 2 | 0 | 0 | 4 | 6 | 0 | 14 | 0 | 1 | 9 | 6 | 7 | 14 | 7 | 10 |
| Replacement | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| X | V | Z | T | L | O | L | X | W | D | B | O | V | L | R | Z | V | Z | Z | C | O | B | J | C | B | R | J | L | P | R | B | X | W | R | X | Y | L | U |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| X | W | Y | H | V | X | M | L | R | A | P | Z | E | D | A | Y | E | U | X | J | . | X | R | ' | U | V | X | C | L | R | A | X | U | M | E | W | W | H |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | . | | | ' | | | | | | | | | | | | | | |

| I | Z | J | E | Y | L | W | R | B | R | X | Z | W | K | P | Z | J | L | U | U | X | U | Z | Y | L | R | X | Y | L | U | P | L | D | P | L | R | O | E | R |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| Y | Z | U | R | V | H | B | K | K | P | L | J | X | B | R | L | . |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | . |

**20. (193.00 pts)** Decrypt the following cipher text that has been encoded using the Porta Cipher using a keyword of STRAIN. What does it say?

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| C | F | M | J | K | B | U | Z | F | B | C | U | N | U | Q | R | E | X | Q | F | B | Z | D | V | Q | R | L | U | D | L | K | J | L | B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| B | J | I | W | Q | A | R | H | W | E | S | Y | V |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | |

**21. (242.00 pts)** A quote has been encoded using the Complete Columnar Cipher for you to decode. The quote has IMPRO somewhere in it. What does it say?

REEHO PSDLG APMOT TATLA EUMED OERIU EBDIV TNVED

ODYAO NSABI HILRL

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| R | E | E | H | O | P | S | D | L | G | A | P | M | O | T | T | A | T | L | A | E | U | M | E | D | O | E | R | I | U | E | B | D | I | V |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| T | N | V | E | D | O | D | Y | A | O | N | S | A | B | I | H | I | L | R | L |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | |

**22. (231.00 pts)** A phrase has been encoded using the Cryptarithm Cipher. What do the values 102487 3802 decode to?

ECHO+SHELL=FROZEN

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| A | A | A | A | A | A | A | A | A | A |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

**23. (344.00 pts)**
[SPECIAL BONUS] A quote has been encoded using the Fractionated Morse Cipher for you to decode. You are told that the quote has PLACES in it corresponding to the encoded text P F E A K E Q O, meaning A = •×•; E = •–•; F = –•×; K = –×–; O = •••; P = ×•–; Q = ×•×. What does it say?

F A I P S V F C M T V E Q P I B M N Z K I E Q P F E A K E Q O Z W L D N I F G J V A F

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A | A |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

**24. (352.00 pts)** A phrase has been encoded using the Aristocrat Cipher. What does it say?

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 0 | 22 | 9 | 15 | 8 | 6 | 1 | 2 | 5 | 7 | 4 | 6 | 6 | 0 | 0 | 6 | 7 | 0 | 2 | 1 | 2 | 2 | 0 | 4 | 3 | 0 | 2 |
| Replacement | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| D | C |
|---|---|
| | |

| W | B | X | B | I | C | T |
|---|---|---|---|---|---|---|
| | | | | | | |

| M |
|---|
| |

| P | C | U | T | I | B | D | B |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

| U | O | E | W | : |
|---|---|---|---|---|
| | | | | : |

| J | D | H | W | K |
|---|---|---|---|---|
| | | | | |

| D | F | B |
|---|---|---|
| | | |

| J | P | O | B | E | P | B |
|---|---|---|---|---|---|---|
| | | | | | | |

| C | Z |
|---|---|
| | |

| M | L | D | ; |
|---|---|---|---|
| | | | ; |

| J | D | H | W | K |
|---|---|---|---|---|
| | | | | |

| D | F | B |
|---|---|---|
| | | |

| M | L | D |
|---|---|---|
| | | |

| C | Z |
|---|---|
| | |

| J | P | O | B | E | P | B | . |
|---|---|---|---|---|---|---|---|
| | | | | | | | . |

| I | B | M | L | E |
|---|---|---|---|---|
| | | | | |

| F | C | G |
|---|---|---|
| | | |

| D | C |
|---|---|
| | |

| J | B | B | . |
|---|---|---|---|
| | | | . |

| L | B | M | I | O | S | B |
|---|---|---|---|---|---|---|
| | | | | | | |

| D | F | M | D |
|---|---|---|---|
| | | | |

| B | X | B | L | K | D | F | O | E | R |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

| P | C | E | E | B | P | D | J |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

| D | C |
|---|---|
| | |

| B | X | B | L | K | D | F | O | E | R |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | |

| B | I | J | B | . |
|---|---|---|---|---|
| | | | | . |

**25. (277.00 pts)**
[SPECIAL BONUS] A message encrypted with the Checkerboard cipher has been received. The 19th through 25th cipher units (MS NO MI ES MO OI ON) decode to be PERSONA. What does it say?

NO EI OS NO YS YS EO NO EI NO OI OC OO ES OO MN ES OC MS NO MI ES MO OI ON EO OS MN ES OO OI NO ES ES MC MO ES OC ES

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| A | A | A | A | A | A | A | A | A | A | A | A |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | |

| A | A |
|---|---|
| | |

| A | A | A | A |
|---|---|---|---|
| | | | |

| A | A | A | A | A | A | A | A |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

| A | A | A | A | A | A | A | A |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

| A | A | A | A | A |
|---|---|---|---|---|
| | | | | |

**26. (371.00 pts)** Decrypt the following cipher text that has been encoded using the Patristocrat Cipher. What does it say?

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency | 0 | 9 | 2 | 5 | 3 | 0 | 10 | 4 | 4 | 10 | 1 | 0 | 0 | 1 | 0 | 1 | 6 | 3 | 0 | 1 | 2 | 3 | 0 | 7 | 1 | 0 | 6 |
| Replacement | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Type your letters below the corresponding given letters. Your letters can be lower or upper case.

| J | K | E | J | I | G | B | S | D | P | U | Z | G | B | N | G | B | G | J | Z | J | I | Z | W | B | G | X | Z | U | D | Z | P | B | H | I |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| W | Q | P | B | D | J | W | T | G | H | G | J | C | P | B | H | G | Q | B | E | J | D | W | U | Z | P | O | E | W | J | I | W | Q | P | B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| D | J | W | T | G | H | G | J | C |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |

**27. (181.00 pts)**
A famous phrase has been encoded as a Baconian. What does it say?

```
* _*** ***__ * _* _* __** * _* __ _ _ _
```

# Archimedean Invitational - Repaired practice edition

### Division B | Codebusters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

**Name: _____ Date: _____**

**2. (200 points) Authored replacement for absent puzzle 2. Decode this Caesar ciphertext with encryption shift +3: PHDVXUH HDFK VBPERO. Use A=0 through Z=25.**

Encryption replaces each plaintext letter with the letter three positions later, wrapping after Z. Spaces are preserved.

Answer: _____
_____
_____

**repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: Cipher: Caesar with forward encryption shift 3. Ciphertext: FDW. Alphabet: A=0 through Z=25.**

Answer: _____
_____
_____